

Student-*t*

Parametrization

The Student-*t* likelihood is defined so that

$$\sqrt{s} \tau (y - \eta) \sim T_\nu$$

for continuous response y where

τ : is the precision parameter

s : is a fixed scaling $s > 0$

η : is the linear predictor

T_ν : is a **reparameterized standard** Student-*t* with $\nu > 2$ degrees of freedom with **unit variance for all values of ν . Please see the example for details!**

Link-function

Identity

Hyperparameters

This likelihood has two hyperparameters

$$\begin{aligned}\theta_1 &= \log(\tau) \\ \theta_2 &= \log(\nu - 2)\end{aligned}$$

and the prior is defined on $\theta = (\theta_1, \theta_2)$.

Specification

- family="T"
- Required argument: y and s (keyword **scale**, default to 1).

Hyperparameter specification and default values

`doc Student-t likelihood`

`hyper`

`theta1`

`hyperid 100001`

`name log precision`

`short.name prec`

`output.name precision for the student-t observations`

`output.name.intern log precision for the student-t observations`

`initial 0`

`fixed FALSE`

`prior loggamma`

`param 1 5e-05`

`to.theta function(x) log(x)`

```

    from.theta function(x) exp(x)
theta2
  hyperid 100002
  name log degrees of freedom
  short.name dof
  output.name degrees of freedom for student-t
  output.name.intern dof_intern for student-t
  initial 5
  fixed FALSE
  prior pc.dof
  param 15 0.5
  to.theta function(x) log(x - 2)
  from.theta function(x) 2 + exp(x)

survival FALSE

discrete FALSE

link default identity

pdf student-t

```

Example

```

#simulate data
n=100
phi=0.85
mu=0.5
eta=rep(0,n)
for(i in 2:n)
  eta[i]=mu+phi*(eta[i-1]-mu)+rnorm(1)
nu=3
t=rt(n,df=nu)
y=eta+t/(sqrt(nu/(nu-2)))
data=list(y=y,z=seq(1:n))
#define the model and fit
formula=y~f(z,model="ar1")
result=inla(formula,family="T",data=data)

```

Notes

None